Extra Practice Questions: Solution Stoichiometry

Complete the following stoichiometric problems. Communicate your problem-solving approach using internationally accepted symbols for elements, quantities, numbers, and units.

1. What is the concentration of a $KOH_{(aq)}$ solution if 12.8 mL of this solution is required to react with 25.0 mL of 0.110 mol/L $H_2SO_{4(aq)}$?

2. What volume of 0.125 mol/L $NaOH_{(aq)}$ is required to react completely with 15.0 mL of 0.100 mol/L $Al_2(SO_4)_{3(aq)}$?

3. In a chemical analysis, a 10.0 mL sample of $H_3PO_{4(aq)}$ was reacted with 18.2 mL of 0.259 mol/L $NaOH_{(aq)}$. Calculate the concentration of the phosphoric acid.

4. The concentration of magnesium ions (assume magnesium chloride) in sea water was analyzed and found to be 50.0 mmol/L. What volume of 0.200 mol/L sodium hydroxide solution would be needed in an industrial process to precipitate all of the magnesium ions from 1.00 ML of sea water?

NOTE
(ML = 1000 000L)
(mmol = 0.001 mol)